# HR Attrition Prediction & Analytics Dashboard


## Abstract

Employee attrition is a major challenge for organizations because high employee turnover increases recruitment costs, training expenses, and productivity loss. The objective of this project is to analyze employee-related factors and predict whether an employee is likely to leave the organization. Machine Learning techniques were applied to historical HR data to identify key factors influencing attrition. An interactive Streamlit dashboard was developed to provide analytics, visualizations, and real-time attrition prediction.


## 1. Introduction

Human Resource Analytics plays a vital role in understanding employee behavior and improving workforce retention. Employee attrition refers to the loss of employees through resignation, retirement, or termination. Predicting employee attrition helps organizations proactively identify employees at risk of leaving and take preventive actions.

This project combines data analysis, machine learning, and dashboard development to build a complete HR Attrition Prediction System.

## 2. Problem Statement

Organizations often struggle to identify employees who are likely to leave the company. Traditional methods rely on manual analysis and may fail to detect hidden patterns in employee data.

The goal of this project is to:

- Analyze HR employee data.
- Identify factors affecting employee attrition.
- Build a predictive machine learning model.
- Provide real-time predictions through an interactive dashboard.

## 3. Objectives

- Perform data cleaning and preprocessing.
- Conduct exploratory data analysis (EDA).
- Identify important factors influencing attrition.
- Compare multiple machine learning models.
- Select the best-performing model.
- Build a Streamlit dashboard for prediction and analytics.
- Store prediction records using SQLite.

## 4. Dataset Description

The project uses the HR Employee Attrition dataset containing employee-related information such as:

- Age

- Gender

- Education

- Job Role

- Job Level

- Monthly Income

- Daily Rate

- Distance From Home

- Overtime

- Work-Life Balance

- Job Satisfaction

- Environment Satisfaction

- Total Working Years

- Years at Company

- Attrition Status

The dataset contains demographic, professional, and satisfaction-related attributes that influence employee retention.

# 5. Data Preprocessing

The following preprocessing steps were performed:

### Missing Value Handling

- Checked for missing values.
- Cleaned and prepared the dataset.

### Categorical Encoding

Categorical features were converted into numerical format using One-Hot Encoding.

Examples:

- Business Travel
- Department
- Education Field
- Job Role
- Marital Status

### Feature Selection

Relevant features were selected for model training.

### Feature Scaling

StandardScaler was applied before training the Support Vector Classifier (SVC) model.

# 6. Exploratory Data Analysis (EDA)

Several visualizations were created to understand employee behavior.

### Visualizations

- Attrition Distribution
- Correlation Heatmap
- Overtime vs Attrition
- Job Satisfaction Analysis
- Monthly Income Distribution
- Department-wise Attrition
- Age Distribution

### Key Findings

    - Employees working overtime showed higher attrition rates.

- Frequent business travel increased attrition probability.
- Single employees were more likely to leave.
- Low job satisfaction contributed significantly to attrition
- Employees with fewer years under the current manager had higher turnover rates.

# 7. Machine Learning Models

## Multiple classification algorithms were evaluated:

### Logistic Regression

- Used as a baseline classification model.

### Decision Tree Classifier

- Provides interpretability and feature importance analysis.

### Random Forest Classifier

- Ensemble learning model with improved generalization.

### Support Vector Classifier (SVC)

- Provides robust classification performance for high-dimensional datasets.

# 8. Model Comparison

The performance of multiple algorithms was compared using classification accuracy.

Model| Performance

Logistic Regression| Evaluated

Decision Tree| Evaluated

Random Forest| Evaluated

Support Vector Classifier| Best Performance

## The Support Vector Classifier achieved the highest accuracy and was selected as the final model.

### Final Model Accuracy

- ✓ Accuracy Achieved: 88%

## 9. Feature Importance Analysis

Permutation Importance was used to identify the most influential features.

Top Factors Affecting Attrition

1. OverTime
2. BusinessTravel_Travel_Frequently
3. MaritalStatus_Single
4. YearsWithCurrManager
5. JobLevel
6. JobSatisfaction
7. TotalWorkingYears
8. DistanceFromHome
9. YearsInCurrentRole
10. DailyRate

These factors significantly contribute to employee attrition behavior

## 10. Dashboard Development

### A Streamlit dashboard was developed to provide:

- Dashboard Features
  - Employee Attrition Prediction
  - HR Analytics Dashboard
  - Interactive Visualizations
  - KPI Metrics
  - Feature Importance Analysis
  - Prediction History
  - Downloadable Results

- Technologies Used
  - Streamlit
  - Plotly
  - Pandas
  - Scikit-Learn
  - SQLite

# 11. Database Integration

SQLite was used to store prediction records and maintain historical data.

Stored Information

- Employee ID

- Prediction Result

- Timestamp

- User Inputs

This enables future analysis and tracking of prediction history.

# 12. Tools and Technologies

**Programming Language:**

- Python

**Libraries:**

- Pandas
- NumPy
- Matplotlib
- Seaborn
- Plotly
- Scikit-Learn
- Joblib
- Streamlit

**Database:**

- SQLite

**Development Environment:**

- Jupyter Notebook
- VS Code

**Version Control:**

- Git
- GitHub

## 13. Results

The developed system successfully predicts employee attrition with an accuracy of 88%.

The analysis identified important employee-related factors that influence attrition and provided actionable insights for HR departments.

The Streamlit dashboard enables real-time prediction and interactive data exploration.

## 14. Conclusion

This project demonstrates the application of Machine Learning and Data Analytics in Human Resource Management. By analyzing employee data and predicting attrition, organizations can proactively identify employees at risk of leaving and implement retention strategies.

The Support Vector Classifier achieved the highest performance and was integrated into an interactive dashboard, providing a complete end-to-end HR Analytics solution.

## 15. Future Enhancements

- Deploy the application on the cloud.
- Add real-time HR data integration.
- Implement advanced ensemble models.
- Add employee retention recommendations.
- Develop a role-based access system.
- Integrate Generative AI for HR insights and reporting.

## References

1. Scikit-Learn Documentation
2. Streamlit Documentation
3. Pandas Documentation
4. IBM HR Employee Attrition Dataset
5. Python Official Documentation